

AI-DND

SAVE_AND_WORLD_STATE_SYSTEM.md

Persistent World State System

Version 0.1

1. DESIGN PHILOSOPHY

The game world never resets.

The world remembers:

- people;
- events;
- wars;
- rumors;
- promises;
- deaths;
- discoveries;
- player actions.

The save file is not a character save.

The save file is a world save.

2. SAVE STRUCTURE

The world state consists of:

WORLD |—— Player |—— NPCs |—— Settlements |—— Factions |—— Economy |—— Diseases
|—— Rumors |—— Wars |—— Quests |—— History |—— Weather |—— Trade |—— World Events
|—— AI Memory |—— Stone Sleep

3. PLAYER SAVE DATA

Player data includes:

- identity;
 - attributes;
 - skills;
 - inventory;
 - reputation;
 - memories;
 - injuries;
 - diseases;
 - fears;
 - relationships;
 - location.
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4. NPC SAVE DATA

Every NPC stores:

- identity;
 - schedule;
 - memories;
 - opinions;
 - relationships;
 - injuries;
 - diseases;
 - goals;
 - emotions;
 - location.
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5. SETTLEMENT DATA

Settlements store:

- population;
- economy;
- diseases;
- reputation;
- buildings;
- factions;
- food supply;

- security.
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6. FACTION DATA

Factions store:

- members;
 - territory;
 - economy;
 - enemies;
 - allies;
 - wars;
 - goals.
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7. RUMOR DATABASE

Rumors store:

- source;
 - location;
 - confidence;
 - witnesses;
 - spread level;
 - age.
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8. HISTORY DATABASE

Historical events store:

- date;
- participants;
- consequences;
- importance.

Examples:

- wars;
 - disasters;
 - heroes;
 - crimes.
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9. WORLD EVENTS

Events store:

- start date;
- end date;
- participants;
- status;
- consequences.

Examples:

- plague;
 - rebellion;
 - invasion.
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10. NPC MEMORY STORAGE

Every memory contains:

- event;
 - emotional impact;
 - participants;
 - confidence;
 - importance;
 - expiration.
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11. RELATIONSHIP STORAGE

Relationships store:

- trust;
 - fear;
 - respect;
 - hatred;
 - gratitude;
 - attraction.
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12. QUEST STORAGE

Quests store:

- creator;
 - participants;
 - objectives;
 - status;
 - consequences;
 - expiration.
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13. WORLD TIMELINE

The game stores:

- year;
- month;
- day;
- hour;
- season.

Nothing stops time.

14. DEATH REGISTRY

The game stores:

- who died;
- when;
- where;
- why;
- witnesses.

Dead NPCs remain in history forever.

15. BIRTH REGISTRY

The game stores:

- births;

- parents;
 - settlements;
 - inheritance.
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16. STONE SLEEP STORAGE

Store:

- sleep start;
 - sleep end;
 - elapsed world time;
 - generated events.
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17. WAR STORAGE

Wars store:

- factions;
 - casualties;
 - territory;
 - economy;
 - leaders.
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18. ECONOMIC STORAGE

Store:

- prices;
 - shortages;
 - production;
 - trade routes.
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19. HEALTH STORAGE

Store:

- diseases;
- epidemics;

- injuries;
 - mortality.
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20. WEATHER STORAGE

Store:

- climate;
 - season;
 - storms;
 - disasters.
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21. MONSTER STORAGE

Store:

- populations;
 - migration;
 - territories;
 - nests.
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22. PLAYER HISTORY

Store:

- promises;
 - crimes;
 - victories;
 - defeats;
 - discoveries.
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23. LEGEND STORAGE

Store:

- rumors;
- myths;
- titles;

- historical reputation.
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24. MEMORY OPTIMIZATION

Important events:

never deleted.

Minor events:

compressed.

Irrelevant events:

removed.

25. SIMULATION LEVELS

Near player:

full simulation.

Far regions:

abstract simulation.

26. SAVE FREQUENCY

Automatic saves occur:

- after combat;
 - after dialogue;
 - after travel;
 - after sleep;
 - after world events.
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27. BACKUP SYSTEM

The game keeps:

- current save;
 - previous save;
 - emergency save.
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28. WORLD CONSISTENCY

Every saved fact must remain consistent.

Example:

Dead NPC:

cannot appear alive.

Destroyed village:

cannot continue normal activity.

29. AI ACCESS

AI systems receive:

- only relevant data;
 - only known information;
 - only accessible memories.
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30. CORE RULE

The save file is not a save.

It is the memory of the world.